

PhD Candidate at Penn State specializing in ML-hardware co-design and AI-driven IC optimization. Researching robust neural architectures and accelerator designs that mitigate circuit non-idealities through bio-inspired plasticity and advanced cross-layer simulations for high-performance computing.

Education

- 08/2022–05/2027 **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, CGPA: 4.00/4.00
(Expected) Advisor: Dr. Abhronil Sengupta
- 08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, CGPA: 4.00/4.00 | Thesis: Neuromorphic Computing for Lifelong Learning
- 02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, CGPA: 3.81/4.00

Academic Research and Teaching Experience

- 08/2022–Present **Graduate Research Assistant**, *Penn State*, University Park, PA
- Conducted research on AI/ML methods for accelerating IC design; developed robust algorithms for specialized AI hardware; integrated ML-driven design flows with industry-standard CAD tools for RTL-level PPA evaluation.
 - Directed characterization of hardware-algorithm interactions; built compact models in Verilog-A and cross-layer simulation frameworks using Python and MATLAB for specialized accelerators.
 - Designed bit-serial accelerators for Transformer workloads; implemented CIM architectures with custom dataflow optimization and non-volatile memory integration.
- 08/2024–05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
 - Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.
- 02/2021–08/2022 **Lecturer**, *ULAB*, Dhaka, Bangladesh
- 08/2022 Taught Solid State Devices, Digital Circuit Design, and Semiconductor Device Physics.

Technical Skills

- Digital/ASIC** Logic synthesis, Digital simulation (Gate/transistor-level), Processor architecture, RTL design, RTL PPA evaluation, Verilog/SystemVerilog, FPGA prototyping, ASIC design flow, Memory/SRAM design, Computer architecture
- ML Systems & Accel.** AI/ML-driven IC design algorithms; Robustness to hardware non-idealities; Circuit noise/mismatch aware ML; AI accelerator research; inference engine design; Systolic Arrays; Transformer/LLM; model compression (PTQ/QAT, pruning, distillation); dataflow optimization; hardware-ML co-design; latency-energy benchmarking
- Hardware & EDA** VLSI design automation, Cadence Virtuoso, Spectre, HSPICE, TCAD, ModelSim, Vivado, CMOS/FinFET, Memory array design, Circuit characterization, Spintronic/Neuromorphic characterization
- Programming** Python, CUDA, C/C++, Linux CLI, Bash/Shell, Docker, performance modeling, architectural simulation, parallel computing
- AI Frameworks** PyTorch, TensorFlow, JAX, ONNX, TensorRT; model development and debugging; inference performance tuning; data tooling (Pandas, NumPy, Jupyter), experiment tracking (W&B, Matplotlib)
- DTCO & STCO** Design-technology and system-technology co-optimization, cross-layer PPA, process-aware design, system-architecture codesign
- Device & Process** Spintronic/Neuromorphic hardware, FeFETs, advanced memory, process integration, TCAD modeling, device-circuit co-design, characterization (DC/AC, IV/CV, AFM, SEM), metrology

AI Dev. Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
Collaboration Git, Research Leadership, Technical Communication, Project Management

Industrial Experience

- 05/2025–07/2025 **Graduate Technical Intern, Intel Corporation, Hillsboro, OR**
- Designed comprehensive Design of Experiments (DOE) for thin film deposition optimization supporting high-volume semiconductor manufacturing.
 - Implemented AI-powered models for automated recipe optimization and real-time monitoring in manufacturing environment.
 - Developed statistical analysis frameworks to assess variability and establish robust control limits for production scalability.
- 09/2020–07/2021 **R&D Engineer, SEMWAVES Ltd., London, UK, part-time**
- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
 - Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Projects

- ML-Driven IC Design Acceleration** Lead Student Researcher, 2023–Present. Developed AI/ML methods to accelerate IC design via automated PPA evaluation and cross-layer simulation. Participated in the development of AI/ML-driven design algorithms robust to hardware mismatch and failures in CIM platforms.
- Astromorphic Transformer** Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture robust to circuit non-idealities through bio-inspired plasticity. Achieved 33.8 perplexity on WikiText-2 and improved convergence on image classification. See: [IEEE TCDS 2025].
- MIPS Micro-processor Design** Lead Student Researcher, 2018–2019. Designed a 5-stage pipelined processor in Verilog; verified functionality through simulation and synthesized for FPGA. Demonstrated core knowledge of processor architectures and digital design flows.
- Neuromorphic Cybersecurity** Lead Student Researcher, 2023–2025. Developed Hierarchical D-SNN for adaptive intrusion detection; achieved 94.3% accuracy. Focused on predicting system behaviors and adapting to dynamic conditions with minimal supervision. See: [ICONS 2025].

Publications

- Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. “Trilinear Compute-in-Memory Architecture for Energy-Efficient Transformer Acceleration”. In: *arXiv preprint arXiv:2604.07628* (2026).
- Md Zesun Ahmed Mia**, Malyaban Bal, Sen Lu, George M. Nishibuchi, Suhas Chelian, Srini Vasan, and Abhronil Sengupta. “Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning”. In: *Proceedings of the International Conference on Neuromorphic Systems (ICONS '25)*. IEEE Press, 2025, pp. 235–238. DOI: 10.1109/ICONS69015.2025.00043.
- Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. “Delving deeper into astromorphic transformers”. In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).
- Arnob Saha, **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. “Toward Variation-Tolerant Ferroelectric Neural Computing: Special Session Paper”. In: *2025 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2025, pp. 583–587.
- KM Ashraful Hoque Fahim, Md Jubair Hasan Khalid, **Md Zesun Ahmed Mia**, and Mirza Rasheduzzaman. “Study of 3-nm Cylindrical GAAFETs with Variations in High-k Dielectric Gate-oxide Materials”. In: *2022 IEEE Symposium on Industrial Electronics & Applications (ISIEA)*. IEEE, 2022, pp. 1–5.

Recognitions

- Melvin P. Bloom Memorial Outstanding Doctoral Research Award (2026)
- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024), Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)
- Student Member, IEEE (2015–Present), Executive Member, EDS, IEEE BD (2021–2022)